Genus

Fusobacteria/Fusobacteriia/Fusobacteriales/Fusobacteriaceae/

Fusobacterium

Knorr 1922, 4[AL]

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Fu.so.bac.te'ri.um. L. n. *fusus* a spindle; L. neut. n. *bacterium* a small rod; N.L. neut. n. *Fusobacterium* a small spindle-shaped rod.

Nonsporeforming rods that are **Gram-stain-negative** and **obligately anaerobic**. Metabolize peptone or carbohydrates in PY-glucose to **produce butyrate**, often with acetate and lower levels of lactate, propionate, succinate, and formate.

DNA G+C content (*mol%*): 26–34.

Type species: **Fusobacterium nucleatum** Knorr 1922, 17[AL] [*Bacillus fusiformis* Veillon and Zuber 1898, 540 and other combinations using "*Fusiformis*" except the organism described as *Fusobacterium fusiforme* by Hoffman in the 7th edition of the *Manual*; Group I, Spaulding and Rettger 1937, 535; Group III (and probably *Fusobacterium polymorphum*) Baird-Parker 1960, 458; *not Fusobacterium plauti-vincentii* Knorr 1922, 5.].

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Number of validated species: 14

Further descriptive information

The genus *Fusobacterium* includes species that do not ferment adonitol, arabinose, dulcitol, glycerol, glycogen, inocitol, inulin, mannitol, melezitose, rhamnose, ribose, sorbitol, or sorbose. Cellobiose is not fermented and esculin is not hydrolyzed except by *Fusobacterium mortiferum* and *Fusobacterium necrogenes*; does not reduce nitrate (except for *Fusobacterium ulcerans*); and does not produce catalase, lecithinase, or acetylmethycarbinol. In addition to butyric, propionic, and acetic acids, species produce variable amounts of butanol from PY medium. Small amounts of formate, lactate, succinate, and ethanol may be produced. Some species convert threonine or lactate to propionate. Pyruvate is converted to acetate and butyrate and sometimes also to formate, succinate, and lactate. H_2S is produced. All species, except *Fusobacterium naviforme* and *Fusobacterium russii*, produce propionate from threonine. Lactate is converted to propionate by *Fusobacterium necroforum* and *Fusobacterium equinum*. Fusobacteria form volatile sulfur compounds from cysteine and methionine (Claesson et al., 1990; Pianotti

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et al., 1986). All species produce indole except for *Fusobacterium mortiferum*, *Fusobacterium necrogenes*, *Fusobacterium russii*, *Fusobacterium ulcerans*, and some strains of *Fusobacterium varium*. Apart from a weak reaction by *Fusobacterium necrophorum*, only *Fusobacterium equinum* produces esterases, while *Fusobacterium canifelinum* is the only fluoroquinolone resistant species of the genus. Additional features are described in Table 1.

Cell morphology

The cells are pleomorphic, some are arranged into filaments, and are spindle-shaped with pointed ends (fusiform) in a few species, while others are coccobacilli. Width is variable. The cells may be single, in pairs end-to-end, or form long coiled filaments. Staining may be irregular and spheroplasts are common in some species. The cells of *Fusobacterium nucleatum* are slender, spindle-shaped with tapered or pointed ends 0.4–0.7 μm thick and 4–10 μm long, and appear singly, in tandem pairs, or in bundles of roughly parallel bacilli. Filaments are often seen in old cultures of *Fusobacterium periodonticum* and *Fusobacterium simiae* which have similar cellular morphology to *Fusobacterium nucleatum*. The cells of *Fusobacterium necrophorum* are pleomorphic, often curved, with rounded and sometimes tapered ends, and they may have spherical enlargements. Free coccoid bodies and especially filaments are common. Cell length may vary from coccobacilli to long threads in clinical samples. *Fusobacterium naviforme* strains have boat-shaped cells. Gonidial forms may be seen in old cultures of *Fusobacterium gonidiaformans*. *Fusobacterium varium* is a small bacillus that does not form filaments. Strains of *Fusobacterium mortiferum* are extremely pleomorphic with globular forms, swellings, and threads. Short rods are predominant in cultures of *Fusobacterium equinum*. The other fusobacteria are pleomorphic filamentous organisms.

Cell-wall composition

Fusobacteria have a cell-wall structure based on two membranes separated by a periplasmic space containing a peptidoglycan layer. *Meso*-lanthioine replaces diaminopimelic acid in *Fusobacterium nucleatum*, *Fusobacterium gonidiaformans*, *Fusobacterium necrophorum*, *Fusobacterium russii*, *Fusobacterium necrogenes*, *Fusobacterium simiae*, and *Fusobacterium periodonticum* (Gharbia and Shah, 1990; Kato et al., 1981; Vasstrand et al., 1982), while *Fusobacterium varium*, *Fusobacterium naviforme*, and *Fusobacterium ulcerans* retain *meso*-diaminopimelic acid and *Fusobacterium mortiferum* contain both versions. *Fusobacterium equinium* and *Fusobacterium canifelinum* have not been tested. Information on the fatty acid content and cell-wall composition of various *Fusobacterium* species can be found in Hofstad (1979), Kato et al. (1981, 1979), Miyagawa et al. (1979), Hofstad and Skaug (1980), Jantzen and Hofstad (1981), and Vasstrand (1981). LPS exhibits O-antigenic specificity linked to lipid A through 2-keto-3-deoxyoctonate. The lipid A of *Fusobacterium nucleatum* is structurally similar to that of *Enterobactereaceae* and cross-reacts serologically with antibodies to *Escherichia coli* lipid A (Dahlen and Mattsby-Baltzer, 1983). *Fusobacterium necrophorum* isolates have a rough-type LPS, whereas that of *Fusobacterium necrophorum* subsp. *necrophorum* is of a smooth type (Brown et al., 1997). The repeat unit of the O-antigenic polysaccharide of *Fusobacterium necrophorum* LPS is an acid identified as a 2-amino-2-deoxy-2-C-methyl-pentonic acid (2-amino-2-methyl-3,4,5-trihydroxypentanoic acid), a novel structure not found before in nature (Hermansson et al., 1993). The main protein of the outer membrane of *Fusobacterium nucleatum*, designated FomA, has a molecular mass of 40 kDa (Bakken et al., 1989b). It is a nonspecific porin present in the outer membrane as a trimer (Kleivdal et al., 1995). The gene encoding the FomA porin has been sequenced (Bolstad et al., 1994). The deduced topology is similar to that of porin structures from other bacteria. *fomA* has been cloned and expressed in *Escherichia coli* (Haake and Wang, 1997; Jensen et al., 1996). Several other outer-membrane proteins have been predicted from the genome sequence, among them some very-high-molecular-mass-proteins (Kapatral et al., 2002). FomA and outer-membrane proteins with molecular masses of 55, 60 and 70 kDa are the major antigens in *Fusobacterium nucleatum* (Bakken et al., 1989a).

The cellular fatty acids in the *Fusobacterium* species examined are straight-chain saturated and monoenoic acids of chain lengths C_{12}-C_{18} and *O*-3-hydroxy-tetradecanoate is distinctive to the oral species *Fusobacterium nucleatum*, *Fusobacterium simiae*, and *Fusobacterium periodonticum* (Jantzen and Hofstad, 1981).

Comparison of small-subunit rRNA sequences has revealed levels of sequence similarity that are consistent with the single genus, but intrageneric heterogeneity is evident (Lawson et al., 1991; Tanner et al., 1994). *Fusobacterium nucleatum*, the species isolated most frequently from humans, is divided into five subspecies; *Fusobacterium nucleatum* subsp. *nucleatum*, subsp. *polymorphum*, subsp. *vincetii*, subsp. *fusiforme*, and subsp. *animalis*, described on the basis of electrophoretic patterns of whole-cell proteins and DNA homology (Dzink et al., 1990) and electrophoretic mobility of two enzymes and DNA homology (Gharbia and Shah, 1992). Recent evidence to support the heterogeneity within *Fusobacterium nucleatum* and the existence of distinct subspecies within

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TABLE 1. Characteristics differentiating the species and subspecies *of Fusobacterium*[a]

Characteristic	*F. nucleatum* subsp. *nucleatum*	*F. nucleatum* subsp. *fusiforme*	*F. nucleatum* subsp. *polymorphum*	*F. nucleatum* subsp. *vincentii*	*F. nucleatum* subsp. *animalis*	*F. canifelinum*	*F. equinum*	*F gonidiaformans*	*F. mortiferum*	*F. naviforme*	*F. necrogenes*	*F. necrophorum* subsp. *necrophorum*	*F. necrophorum* subsp. *fundulíforme*	*F. perfodens*	*F. periodonticum*	*F. russii*	*F. simae*	*F. ulcerans*	*F. varium*
Cellobiose	–	–	–	–	–	–	–	–	w⁻	–	w⁻	–	–	–	–	–	–	–	–
Esculin hydrolysis	–	–	–	–	–	–	–	–	+⁻	–	+	–	–	–	–	–	–	–	–
Utilization of:																			
Fructose	w⁻	w⁻	w⁻	w⁻	w⁻	w⁻	–	–	w[a]	–	w[a]	–[w]	–[w]	w	w⁻	–	w	–	w[a]
Gelatin	–[w]	–[w]	–[w]	–[w]	–[w]	–[w]	–[w]	–	–	–	–	v	v	–	–	–	–	–	–
Glucose	–[w]	–[w]	–[w]	–[w]	–[w]	–[w]	–	w⁻	a[w]	–[w]	w[a]	–[w]	–[w]	w	w⁻	–	w	a[w]	w[a]
Lactose	–	–	–	–	–	–	–	–	w[a]	–	–	–	–	–	–	–	–	–	–
Maltose	–	–	–	–	–	–	–	–	w⁻	–	–	–	–	–	–	–	–	–	–
Mannose	–	–	–	–	–	–	–	–	w[a]	–	w[a]	–	–	–	–	–	–	a[w]	w
Raffinose	–	–	–	–	–	–	–	–	v	–	–[w]	–	–	–	–	–	–	–	–
Sucrose	–	–	–	–	–	–	–	–	v	–	–[w]	–	–	w	–	–	–	–	–
Indole	+	+	+	+	+	+	+	+	–	+	–	+	+	–	+	–	+	–	+⁻
Nitrate	–	–	–	–	–	–		–	–	–	–	–	–	–	–	–	–	+	–
Bile growth	–	–	–	–	+	–	+	–	+	–	+	v	v	–	–	–	+	+	+
Lipase	–	–	–	–	–	–	+	–	–	–	–	+⁻		–	–	–	w	–	–[w]
β-Hemolysis	–	–	–	–	–	–	–	–	–	–	–	+	–	–	–	–	–	–	–
Gas produced in agar	–	–	–	–	–	–	nt	+	+	–	+	+	+	+	–	–	–	+	+
Lactate→propionate	–	–	–	–	–	–	+	–	–	–	–	+	+	–	–	–	–	–	–
Threonine→propionate	+	+	+	+	+	+	+	+	+	–	+	+	+	+	+	–	+	+	+
Activity of:																			
N-Acetyl-glucosaminidase	–	–	–	–	–	–	–	–	–	–	–	–	–	+	–	–	–	–	–
Alkaline phosphatase	–	–	–	–	–	–	–	–	+	–	+	+	–	–	–	w	–	–	w
Acid-phosphatase	–	–	–	–	–	–	+	w	+	–	+	+	+	w	–	+	–	+	w
α-Galactosidase	–	–	–	–	–	–	–	–	+	–	+	–	–	–	–	–	–	–	–
β-Galactosidase	–	–	–	–	–	–	–	–	+	–	+	–	–	+	–	–	–	–	–
β-Glucoronidase	–	–	–	–	–	–	–	–	–	+	–	–	–	–	–	–	–	–	–
α-Glucosidase	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
β-Glucosidase	–	–	–	–	–	–	–	–	+⁻	–	–	–	–	–	–	–	–	–	–
Esterase (C4)	–	–	–	–	–	–	+	–	–	–	–	w	w	–	–	–	w	–	–
Esterase (C8)	–	–	–	–	–	–	+	–	–	–	–	w	w	–	–	–	–	–	–

[a]Symbols: w, weakly positive; w⁻: most strains weakly positive, some negative; –[w]: most strains negative, some weakly positive; a[w]: strongly acid with some weakly acid reactions; w[a]: usually weakly acid with occasional strong acid reactions; nt, not tested. (Adapted from Conrads et al., 2004a.)

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Fusobacterium nucleatum was obtained by comparison of their 16S–23S internal transcribed spacer regions (Conrads et al., 2002).

Fusobacterium periodonticum is phylogenetically similar to *Fusobacterium nucleatum* (Jousimies-Somer and Summanen, 2002). *Fusobacterium nucleatum*-like isolates from cats and dogs were found to be distinct from *Fusobacterium nucleatum* both in their 16S rRNA and DNA–DNA hybridization. These have been reclassified as *Fusobacterium canifelinum* which are resistant to levofloxacin (MIC >4 μg/ml) and other fluoroquinolones. The resistance is due to two substitutions in the quinolone resistance determining region of *gyrA* relative to *Fusobacterium nucleatum*. The first replacement is of Ser79 with leucine and the second is Gly83 is replaced with arginine (Conrads et al., 2005).

Animal isolates of *Fusobacterium necrophorum* form two subspecies: *Fusobacterium necrophorum* subsp. *necrophorum* and *Fusobacterium necrophorum* subsp. *funduliforme* (Shinjo et al., 1991) corresponding to biovars A and B. A distinctive feature separating the subspecies genetically is the presence of isoleucine and alanine tRNA gene in *Fusobacterium necrophorum* subsp. *necroforum*, while *Fusobacterium necrophorum* subsp. *funduliforme* contains the isoleucine tRNA gene (Jin et al., 2002).

Genetic differences between the species have been reported in leukotoxin A (*lktA*) between both subspecies (Narayanan et al., 2001). Subspecies *necrophorum* is more frequently isolated, often in pure culture, from liver abscesses than subsp. *funduliforme*. Leukotoxin, an exotoxin, is a major virulence factor. In *Fusobacterium necrophorum* subsp. *necrophorum*, Lkt is a high-molecular-mass protein that is encoded by a tricistronic leukotoxin operon (*lktBAC*) and induces apoptosis and necrosis of bovine leukocytes in a dose-dependent manner. The subsp. *funduliforme* produces lower concentration of leukotoxin and hence is less virulent than subsp. *necrophorum*. The low toxicity associated with subsp. *funduliforme* leukotoxin, a less virulent subspecies, may in part be due to the differences in the *lktA* gene and reduced transcription (Tadepalli et al., 2008).

Fusobacterium mortiferum, *Fusobacterium varium*, and *Fusobacterium ulcerans* share several phenotypic properties similar to *Fusobacterium varium*, but *Fusobacterium ulcerans* reduces nitrate to nitrite (Citron, 2002). It also contains unique fragments of 1000 and 550 bp not present in *Fusobacterium varium* and *Fusobacterium mortiferum* (Claros et al., 1999). *Fusobacterium necrogenes* is distantly related to the other *Fusobacterium* species. Based on 16S–23S rDNA spacer region sequences (Conrads et al., 2002) *Fusobacterium gonidiaformans* is genealogically related to *Fusobacterium necrophorum* (Nicholson et al., 1994).

Metabolism

The core metabolism of *Fusobacterium nucleatum* is similar to that of *Clostridium*, *Lactococcus*, and *Enterococcus* species (Kapatral et al., 2002). More than 137 transporters for the uptake of substrates such as peptides, sugars, metal ions, and cofactors have been detected. Amino acids and small peptides comprise the major sources of energy for all *Fusobacterium* species (Gharbia and H.N. Shah, 1989), however, peptides influence the uptake of amino acids enhancing the utilization of histidine and glutamate while threonine, methionine, and asparagine were repressed (Gharbia et al., 1989). Acidic and cationic amino acids are the main acids incorporated. Glutamate, histidine, lysine, and serine appear to be key amino acids in *Fusobacterium nucleatum* (Gharbia and Shah, 1991a; Rogers et al., 1992). Biosynthetic pathways exist for glutamate, aspartate, and glutamylglutamate to be used as growth substrates for *Fusobacterium nucleatum* (Takahashi and Sato, 2002). Glutamate is a key catabolic substrate in *Fusobacterium* species (Gharbia and Shah, 1991b). It is catabolized via the 2-oxoglutarate pathway with the production of acetate and butyrate as end products (Gharbia and Shah, 1991b). Glutamate may also be degraded by the methylaspartate pathway in *Fusobacterium varium* (Gharbia and Shah, 1991b). Enzymes representative for the mesaconate pathway for catabolism of glutamate have been detected in *Fusobacterium varium*, *Fusobacterium mortiferum*, and *Fusobacterium ulcerans* (Gharbia and Shah, 1991b). *Fusobacterium varium* and *Fusobacterium mortiferum* also possess enzymes representative of the 4-aminobutyrate pathway.

Amino acids are imported as monomers, di-, or oligopeptides, and an active transport of the dipeptide L-cysteinglycine has been detected in *Fusobacterium nucleatum* (Carlsson et al., 1994).

Fusobacterium species differ in their ability to use fermentable carbohydrates as energy sources for growth (Robrish et al., 1991). *Fusobacterium nucleatum* and other species utilize glucose for biosynthesis of intracellular glycopolymers that can be degraded to produce energy under conditions of amino-acid deprivation (Robrish et al., 1987). The accumulation of glucose is dependent on energy generated by the fermentation of amino acids (Robrish et al., 1987). *Fusobacterium mortiferum* is an exception in that accumulation of sugars is independent of a fermentable amino acid. Significantly, *Fusobacterium mortiferum* has the ability to metabolize various sugars as energy sources for growth (Robrish et al.,

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1991). Sugars utilized include α- and β-glycosides, which are transported by the phosphoenolpyruvate-dependent sugar: phosphotransferase system. Thus, it utilizes sucrose and its isomeric α-D-glucosyl-D-fructoses as energy sources for growth (Pikis et al., 2002).

The genes encoding phospho-β-glucosidase (P-β-glucosidase; EC 3.22.1.86) and 6-phospho-α-D-glucosidase (maltose-6-phosphate hydrolase; EC 3.2.1.122) known as *pbgA* and *malH*, respectively, have been expressed in *Escherichia coli* (Bouma et al., 1997; Thompson et al., 1997).

Genetics

The genome of *Fusobacterium nucleatum* subsp. *nucleatum* (strain ATCC 25586^T) contains 2.17 Mb encoding 2067 open reading frames (ORFs) organized on a single circular chromosome (Kapatral et al., 2002). About 2.3% of the ORFs are unique to *Fusobacterium nucleatum*. The genome analysis has revealed several key aspects of the pathways of organic acid, amino acid, carbohydrate, and lipid metabolism. Nine very-high-molecular-mass outer-membrane proteins are predicted from the sequence, none of which has been reported in the literature. More than 137 transporters for the uptake of a variety of substrates such as peptides, sugars, metal ions, and cofactors have been identified. Biosynthetic pathways exist for only three amino acids: glutamate, aspartate, and asparagine. The remaining amino acids are imported as such or as di- or oligopeptides that are subsequently degraded in the cytoplasm. A principal source of energy appears to be the fermentation of glutamate to butyrate. Additionally, desulfuration of cysteine and methionine yields ammonia, H_2S, methyl mercaptan, and butyrate, which are capable of arresting fibroblast growth, thus preventing wound healing and aiding penetration of the gingival epithelium.

Analysis of the draft genome sequence of *Fusobacterium nucleatum* subsp. *vincentii* (ATCC 49256), and comparison of this sequence to the genome ATCC 25586 sequence resolved that 441 ORFs have no orthologs in strain ATCC 25586. Of these, 118 ORFs have no known function and are unique, whereas 323 ORFs have functional orthologs in other organisms. Genes for eukaryotic serine/threonine kinase and phosphatase, transpeptidase E-transglycosylase Pbp1A were also identified among the ATCC 49256-specific ORFs. Unique ABC transporters, cryptic phages, and three types of restriction-modification systems have been identified in ATCC 49256. ORFs for ethanolamine utilization, thermostable carboxypeptidase, glutamyl-transpeptidase, and deblocking aminopeptidases are absent from ATCC 49256. Both strains lack a catalase-peroxidase system, but possess thioredoxin/glutaredoxin enzymes. Genes for resistance to antibiotics such as acriflavin, bacitracin, bleomycin, daunorubicin, florfenicol, and other general multidrug resistance are present (Kapatral et al., 2003).

The genome of *Fusobacterium nucleatum* subsp. *polymorphum* ATCC 10953 contains a chromosome of approximately 2.4 Mbp and a plasmid (pFN3) of 11.9 kbp, and there are 2361 proteins encoded in the genome. Plasmid pFN3 from the strain was also sequenced and analyzed. When compared to the other two available fusobacterial genomes (*Fusobacterium nucleatum* subsp. *nucleatum* and *Fusobacterium nucleatum* subsp. *vincentii*) 627 ORFs unique to *Fusobacterium nucleatum* subsp. *polymorphum* ATCC 10953 were identified. A large percentage of these mapped within one of 28 regions or islands containing five or more genes. Seventeen percent of the clustered proteins that demonstrated similarity were most similar to proteins from the clostridia, with others being most similar to proteins from other Gram-stain-positive organisms such as *Bacillus* and *Streptococcus*. A 10 kb region homologous to the *Salmonella typhimurium* propanediol utilization locus was identified, as was a prophage and integrated conjugal plasmid. The genome contains five composite ribozyme/transposons similar to the Cd*IS*t IStrons described in *Clostridium difficile*.

Additionally, three plasmids isolated from *Fusobacterium nucleatum* strains were sequenced. These are designated pFN1, pPA52, and pKH9 (Bachrach et al., 2004; Haake et al., 2000; McKay et al., 1995).

Ecology

Fusobacterium are normal inhabitants of the mucous membrane of humans and animals. The habitat of *Fusobacterium nucleatum* and *Fusobacterium periodonticum* is the human oral cavity; the gingival crevice is the niche of both. Strains are isolated from the oral microflora of adults and children and are also found in edentulous infants. The vagina is likely to be the primary endogenous site for *Fusobacterium naviforme* and *Fusobacterium gonidiaformans* (Hill, 1993). The habitat of *Fusobacterium mortiforum* and *Fusobacterium varium* is the gastrointestinal tract, where they are present in small numbers. *Fusobacterium necrogenes*, which was originally isolated from a chicken abscess and from cecal contents of ducks, is very rarely isolated from humans. The habitat of *Fusobacterium ulcerans*, isolated from tropical ulcers, is unknown.

Fusobacterium necrophorum is a normal inhabitant of the alimentary tract of cattle, horses, sheep, and pigs and is frequently isolated from cats and dogs. Reports, mainly from the first part of the twentieth century, indicate its presence in a range of wild animals including reptiles. However, the

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relatively frequent isolation of *Fusobacterium necroforum* from soft-tissue infections of the oral cavity and the upper respiratory tract compared to elsewhere in the body, suggest that these sites are the principal human habitat of this organism.

Fusobacterium russii is normally found in the canine and feline oral flora (Love et al., 1987), but it has also been isolated from human feces. The oral cavity of horses and the stump-tailed macaque (*Macaca arctoides*) are the habitat of *Fusobacterium equinum* and *Fusobacterium simiae*, respectively. *Fusobacterium canifelinum* was isolated from the microflora of infected cat and dog bite wounds in humans (Conrads et al., 2005).

Maintenance procedures

Fusobacteria are not particularly demanding with regard to a low oxidation-reduction potential. They are, however, killed fairly readily by exposure to oxygen. This is possibly due to their susceptibility to peroxides. Growth is best at 35–37°C and at a pH near 7.

All *Fusobacterium* species grow on blood agar based on proteose-peptone, tryptone, or trypticase. Good growth is usually obtained in a rich semifluid medium such as brain heart medium supplemented with yeast extract. Batch cultivation is best performed in a fluid medium containing a reducing agent.

Laboratory isolation and identification

Fusobacterium species can be separated from other related taxa based on their phylogenetic relatedness and distinctive phenotypic characteristics (Table 2).

The isolation of fusobacteria from an aerobic infection requires proper collection of the specimen, the use of an anaerobic transport vial, optimal anaerobiosis (an anaerobic chamber is not necessary), and a rich medium for optimal growth. Fastidious Anaerobe Agar Base (FAA, Lab M) supplemented with 5% sheep or horse blood is recommended for culture.

The use of a selective medium is necessary for the enumeration of viable fusobacteria in specimens of normal flora. The addition of josamycin, vancomycin, and norfloxacin plus 5% defibrinated horse blood to FAA supports the growth of *Fusobacterium* species while completely inhibiting most other obligate anaerobes and facultative bacteria (Brazier et al., 1991). Rifampin blood agar (Sutter et al., 1971) is useful for the selective isolation of *Fusobacterium varium* and *Fusobacterium mortiferum*. Diagnostic tables are convenient for the examination of bile resistance, production of alkaline phosphatase, and orthonitrophenol-β-D-galactopyranoside (ONPG)-test. Commercially available kits based on the detection of preformed enzymes and designed for identification of anaerobes may be of help in supporting a suspected identity of a *Fusobacterium* isolate.

Surface colonies of *Fusobacterium nucleatum* are 1–2 mm in diameter after incubation for 2 d. They are white to yellow-gray in color, speckled, smooth or breadcrumb-like. The colonies are usually nonhemolytic. Cultures of *Fusobacterium nucleatum* produce an unpleasant, but not fetid, smell. After incubation for 2 d, colonies of animal isolates of *Fusobacterium necrophorum* produce a foul, putrid odor. Fluid or semifluid cultures are characterized by abundant production of gas, and colonies are about 2 mm in diameter, flat, circular with scalloped to erose edge, opaque, and white to gray (subsp. *necrophorum*); or 1 mm wide, circular with entire margins, gray, translucent, and with smooth surfaces (subsp. *funduliforme*) (Shinjo et al., 1991).

The colonial morphology of *Fusobacterium necrophorum* isolated from human infections is similar to that of *Fusobacterium necrophorum* subsp. *funduliforme*. The colonies are small and have been described as creamy, opaque, smooth, umbonate or raised, and with entire edges (Hall et al., 1997). The colonial morphology of most other *Fusobacterium* species is similar to that of *Fusobacterium necrophorum* subsp. *funduliforme*.

All *Fusobacterium* species produce butyric acid. They are catalase and nitrate negative, sensitive to kanamycin (1000 μg disk) and colistin (10 μg disk), resistant to vancomycin (5 μg disk), and produce a rancid odor.

Determination of the electrophoretic migration of glutamate dehydrogenase and 2-oxoglutarate reductase provides a rapid method for the identification of most *Fusobacterium* species (Gharbia and Shah, 1990). Identification based on gas-liquid chromatography of cellular fatty acids may be of value in reference laboratories, provided a database is used as a basis for comparison. A DNA probe has been used successfully to detect *Fusobacterium nucleatum* directly in samples of subgingival plaque (Lippke et al., 1991), and checkerboard DNA–DNA hybridization has been used successfully to detect *Fusobacterium nucleatum* and subspecies in periodontal and endodontic infections (Socransky et al., 1998; Sunde et al., 2000). A PCR-detection of part of the major 40 kDa outer-membrane protein, the FomA porin of *Fusobacterium nucleatum*, is specific for *Fusobacterium nucleatum* when used under condition of high stringency (Bolstad and Jensen, 1993). Sequencing of 16S rRNA from clinical isolates provides a discriminatory tool for separating all species (Figure 1).

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TABLE 2. Characteristics differentiating *Fusobacterium* from genetically related taxa[a]

Reaction	*Fusobacterium nucleatum*	*Cetobacterium ceti*	*Leptotrichia buccalis*	*Sebaldella termitidis*	*Sneathia sanguinegens*	*Streptobacillus moniliformis*
Catalase	–	–	–	–	–	–
Oxidase	nd	–	nd	nd	–	–
Indole	+	+	–	–	–	–
Arginine dihydroylase	nd	nd	nd	nd	+	+
Nitrate to nitrite	–	v	–	–	–	–
Esculin hydrolysis	–	–	+	nd	+	v
Gelatinase	–	–	–	–	nd	–
Starch hydrolysis	nd	–	v	–	–	nd
ONPG	nd	+	nd	nd	nd	–
Acid from:						
Arabinose	nd	nd	–	–	–	–
Cellobiose	–	nd	+	nd	nd	–
Fructose	–w	nd	+	+	nd	+
Galactose	nd	nd	d	nd	nd	+
Glucose	–w	nd	+	+	+	+
Glycogen	nd	nd	nd	nd	–	+
Lactose	–	nd	+	–	–	–
Maltose	–	nd	+	+	–	+
Mannose	–	nd	+	nd	v	+
Mannitol	–	nd	–	+	–	–
Starch	–	nd	d	–	–	+
Sucrose	–	nd	+	+	–	–
Trehalose	–	nd	+	+	–	–
Xylose	–	nd	–	+	nd	–

[a]Symbols: +, 90% or more strains are positive; –, 90% or more strains are negative; d, 11–89% of the strains are positive; ONPG, *O*-nitrophenyl-β-D-galactopyranoside; v, variable; w, weak; nd, no data.

List of species of the genus *Fusobacterium*

Fusobacterium nucleatum
Knorr 1922, 17[AL] [*Bacillus fusiformis* Veillon and Zuber 1898, 540 and other combinations using "*Fusiformis*" except the organism described as *Fusobacterium fusiforme* by Hoffman in the 7th edition of the *Manual*; Group I, Spaulding and Rettger 1937, 535; Group III (and probably *Fusobacterium polymorphum*) Baird-Parker 1960, 458; *not Fusobacterium plauti-vincentii* Knorr 1922, 5.]

nu.cle.a'tum. L. neut. adj. *nucleatum* having a kernel, intended to mean nucleated.

Cells from glucose broth cultures are 0.4–0.7 × 3 μm, have tapered to pointed ends, and often have central swellings and intracellular granules. Cell length is variable but is usually fairly uniform within actively growing cultures. Cells do not possess pili or flagella (Dahlen et al., 1978; Falkler and Hawley, 1977). Surface colonies on blood agar are 1–2 mm in diameter, circular to slightly irregular, convex to pulvinate, translucent often with a "flecked" appearance, usually nonhemolytic (horse or rabbit blood), but may be slightly hemolytic under the area of confluent growth or may produce greenish discoloration of the blood agar upon exposure to oxygen. Glucose broth cultures have a flocculent or granular sediment with or without turbidity, a final pH of 5.6–6.2, and a foul "bad breath" odor. Produces

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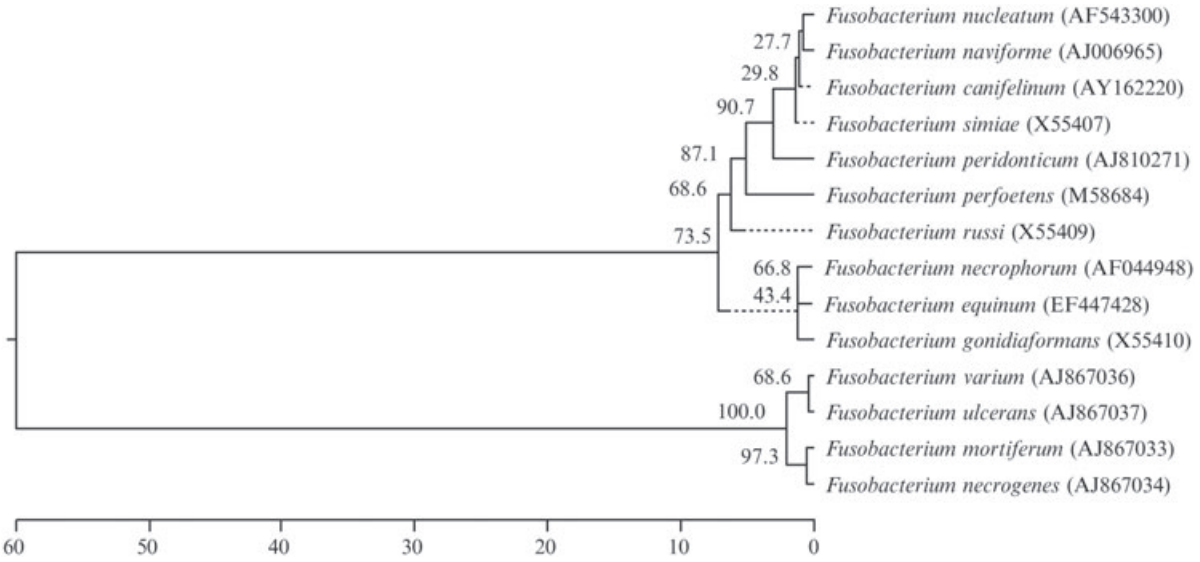
FIGURE 1. Unrooted neighbor-joining tree of the 16S rRNA gene sequences of *Fusobacterium*. GenBank accession numbers are given after species names. The numbers above branches are bootstrap percentages from 1000 resampled datasets. Bar = 0.1 difference per nucleotide.



DNase (Porschen and Sonntag, 1974). No phosphatase detected (Porschen and Sonntag, 1974). Most strains produce H_2S. Capable of hemagglutinating human and animal erythrocytes. Contains the diamino acid lanthionine in cell-wall peptidoglycan as a major component; no lysine, diaminopimelic acid, or ornithine (Gharbia and Shah, 1990; Kato et al., 1979; Vasstrand, 1981). Heptose and KDO are present in the lipopolysaccharide (Hofstad, 1974). Grows in the presence of up to 6% oxygen; survives exposure to air for 100 min (Loesche, 1969).

Strains are resistant to 3 μg/ml of erythromycin (broth disc test). Some *Fusobacterium nucleatum* isolates produce β-lactamase, an activity not reported for other *Fusobacterium* species. A penicillin-hydrolyzing β-lactamase inhibited by clavulanic acid has been isolated from *Fusobacterium nucleatum* (Turner et al., 1985). *Fusobacterium nucleatum*, similar to other species of the genus, are susceptible to amoxycillin/clavulanate, carbapenems, chloramphenicol, quinolone clinafloxalin, linezolide, and nitroimidazoles (Goldstein et al., 1999). Resistance to cefoxitin and clindamycin is very low. Resistance to tetracyclines is common in *Fusobacterium nucleatum* since a high percentage of strains harbor a chromosomal *tetM* locus. Alternatively, a tetM determinant on a conjugal transposon (Roberts and Lansciardi, 1990) was also detected. Resistance to glycylglycines has not been observed (Downes et al., 1999). *Fusobacterium nucleatum* is sensitive to the bactericidal action of protegrins and other antibacterial peptides (Miyasaki et al., 1998). Other characteristics of the species are given in Table 1.

DNA G+C content (mol%): 26–28 (T_m).

Type strain: ATCC 25586, CCUG 32989, CCUG 33059, CIP 101130, JCM 8532, LMG 13131.

Sequence accession no. (16S rRNA gene): AJ133496.

Further comments: *Fusobacterium nucleatum* is divided into five subspecies: *Fusobacterium nucleatum* subsp. *nucleatum*, subsp. *animalis*, subsp. *fusiforme*, subsp. *polymorphum*, and subsp. *vincentii*, described on the basis of electrophoretic patterns of whole-cell proteins and DNA homology (Dzink et al., 1990) and electrophoretic mobility of glutamate dehydrogenase and 2-oxoglutarate dehydrogenases and DNA homology (Gharbia and Shah, 1992). The five subspecies differed from each other and from other closely related species by comparisons of their 16S–23S internal transcribed spacer regions (Conrads et al., 2002).

Fusobacterium nucleatum subsp. nucleatum (Knorr 1922) Dzink, Sheenan and Socransky 1990, 77[VP] (*Fusobacterium nucleatum* Knorr 1922, 17)

nu.cle.a'tum. L. neut. adj. *nucleatum* having a kernel, intended to mean nucleated.

DNA G+C content (mol%): 26–28 (T_m).

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Type strain: ATCC 25586, CCUG 32989, CCUG 33059, CIP 101130, JCM 8532, LMG 13131.

Sequence accession no. (*16S rRNA gene*): AJ133496.

Further information: the subspecies name *Fusobacterium nucleatum* subsp. *nucleatum* Knorr (1922) is automatically created by the valid publication of *Fusobacterium nucleatum* subsp. *polymorphum* (*ex* Knorr (1922) Dzink (1990), and the valid publication of *Fusobacterium nucleatum* subsp. *vincentii* (*ex* Knorr (1922) Dzink (1990) [Rule 40d (formerly Rule 46)].

Fusobacterium nucleatum subsp. animalis Gharbia and Shah 1992, 297[VP]

a.ni.ma'lis. L. n. *animal*, *-alis* an animal; L. gen. n. *animalis* of an animal.

Type strain: ATCC 51191, CCUG 32879, CIP 104879, JCM 11025, NCTC 12276.

Sequence accession no. (*16S rRNA gene*): X55404.

Fusobacterium nucleatum subsp. fusiforme (*ex* Veillon and Zuber 1898) Gharbia and Shah 1992, 297[VP] ("*Sphaerophorus fusiformis*" Veillon and Zuber 1898; Sebald 1962)

fu.si.for'me. L.n. *fusus* a spindle; L. neut. suff. *forme* in shape of; N.L. neut. adj. *fusiforme* spindle-shaped.

Type strain: ATCC 51190, CCUG 32880, CIP 104878, DSM 19508, JCM 11024, NCTC 11326.

Sequence accession no. (*16S rRNA gene*): AM849219.

Fusobacterium nucleatum subsp. polymorphum (Knorr 1922) Dzink, Sheenan and Socransky 1990, 77[VP] ("*Fusobacterium polymorphum*" Knorr 1922)

po.ly.mor'phum. N.L. neut. adj. *polymorphum* (from Gr. adj. *polumorphos -on*) multiform, polymorphic.

Type strain: ATCC 10953, CCUG 9126, DSM 20482, JCM 12990, NCTC 10562.

Sequence accession no. (*16S rRNA gene*): AF287812.

Fusobacterium nucleatum subsp. vincentii (Knorr 1922) Dzink, Sheenan and Socransky 1990, 77[VP] ("*Fusobacterium plauti-vincentii*" Knorr 1922)

vin.cen'.ti.i. N.L. masc. gen. n. *vincentii* of Vincent, referring to Henri Vincent who studied the organism originally isolated from Vincent's angina and necrotizing ulcerative gingivitis.

Type strain: ATCC 49256, CCUG 37843, CIP 104988, DSM 19507, JCM 11023.

Sequence accession no. (*16S rRNA gene*): AJ006964, AM887529.

Fusobacterium canifelinum Conrads, Citron, Mutters, Jang and Goldstein 2004b, 1909[VP] (Effective publication: Conrads, Citron, Mutters, Jang and Goldstein 2004a, 412.)

ca. ni.felí num. L. gen. pl. n. *canum* of dogs; L. neut. adj. *felinum* of or belonging to a cat; N.L. neut. adj. *canifelinum* of dogs and cats.

Strains grown on supplemented Brucella blood-agar for 2 d; colonies are convex, 1–2 mm in diameter with a slightly lobate margin, white, opaque with a granular internal appearance. Cells are slender Gram-stain-negative rods with pointed ends. Glucose and fructose are fermented weakly with a terminal pH 5.7–6.0. End product analysis of PY-glucose by gas-liquid chromatography reveals acetic and butyric acids. Threonine, but not lactate, is converted to propionate. The isolates produce indole, fail to grow in bile, do not hydrolyze esculin, and do not produce acid from lactose, maltose, mannose, raffinose, and sucrose. On agar dilution sensitivity tests, all isolates tested were susceptible to penicillin G, and metronidazole. All strains were resistant (MIC >4 μg/ml) to levofloxacin, moxifloxacin, gemifloxacin, and other fluoroquinolones. *Fusobacterium canifelinum* can be distinguished phenotypically from other species of the genus *Fusobacterium* by the resistance to fluoroquinolones.

Source: a purulent dog-bite wound in a human patient.

DNA G+C content (*mol%*): 26–28 (T_m).

Type strain: RMA 1036, ATCC BAA-689, CCUG 49733, DSM 15542.

Sequence accession no. (*16S rRNA gene*): AY162221.

Fusobacterium equinum Dorsch, Love and Bailey 2001, 1962[VP]

e.quin'um. L neut. adj. *equinum* of horses.

On sheep-blood agar after 2 d, colonies are circular with an entire undulate margin, convex to umbonate, 1–2 mm in diameter, and creamish in color. Gram staining of single colonies on primary plates reveals pleomorphic Gram-stain-negative rods (coccobacilli to rods, with rounded ends, and curved rods often with irregular staining), but on subculture, coccobacilli and short rods predominate. The terminal pH in glucose liquid medium ranges from pH 6.8–7.0. On GLC analysis, acetic, propionic, and butyric acids are produced and lactate and threonine are converted to propionate. The isolates produce indole and grow in bile and are nonmotile and fail to hydrolyze esculin and starch. The isolates do not ferment esculin, fructose, glucose, lactose, maltose, mannose, starch, and sucrose. They do not hemagglutinate chicken erythrocytes. They are lipase and

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lecithinase positive. On broth disc sensitivity tests, all isolates are sensitive to penicillin G, amoxycillin, chloramphenicol, doxycycline, and metronidazole; all strains except VPB 4076 and VPB 4014 are sensitive to erythromycin.

Source: pus obtained from a discharged sinus associated with the pirulent para-oral lesion in a horse.

DNA G+C content (*mol%*): 29–31 (T_m).

Type strain: JCM 11174, NCTC 13176, VPB 4027.

Sequence accession no. (*16S rRNA gene*): AJ295750.

Fusobacterium gonidiaformans (Tunnicliff and Jackson 1925) Moore and Holdeman 1970, 45[AL] [*Bacillus gonidiaformans* Tunnicliff and Jackson 1925, 430; *Sphaerophorus gonidiaformans* (Tunnicliff and Jackson 1925) Prévot 1938, 299]

go.ni.di.a.for'mans. Gr. n. *gone* offspring, seed; N.L. n. *gonidium* gonidium; N.L. pl. n. *gonidia* gonidia; L. part. adj. *formans* forming; N.L. part. adj. *gonidiaformans* gonidia forming.

Cells from glucose broth cultures are pleomorphic and vacuolated, 0.4–0.7 × 0.7–3.0 µm, often with degenerate filaments or long strands. The spheroid or gonidial forms implied by the name of this organism are seen most often in old cultures or in media that are not highly reduced. Surface colonies on horse blood agar plates are punctiform to 1 mm in diameter, circular entire, low convex, translucent, and smooth. Glucose broth cultures are turbid with smooth sediment and a final pH of 5.6–6.2. Produces DNase (Porschen and Sonntag, 1974). No phosphatase is detected (Porschen and Spaulding, 1974). Hippurate is hydrolyzed. Other characteristics of the species are given in Table 1.

Source: the intestinal and urogenital tracts of humans, various types of human infections and from a lamb with pneumonia.

DNA G+C content (*mol%*): 33 (T_m) (Gharbia and Shah, 1990).

Type strain: ATCC 25563, CCUG 16790.

Sequence accession no. (*16S rRNA gene*): X55410.

Fusobacterium mortiferum (Harris 1901) Moore and Holdeman 1970, 45[AL] [*Bacillus mortiferus* Harris 1901, 546; *Sphaerophorus mortiferus* (*sic*) (Harris 1901) Prévot 1938, 299]

mor.ti'fer.um. L. neut. adj. *mortiferum* death-bringing, death-bearing.

Cells from glucose broth cultures are 0.8–1.0 × 1.5–10 µm occurring singly and in pairs and short chains. Cells stain irregularly and may be extremely pleomorphic. Surface colonies on horse blood agar are 1–2 mm in diameter, circular with entire, diffuse, or slightly scalloped edge; convex; translucent; smooth. Glucose broth cultures are uniformly turbid with smooth or semiviscous sediment. No superoxide dismutase is detected (Gregory et al., 1978). Lysine decarboxylase is not produced (Werner, 1974). Produces DNase and phosphatase (Porschen and Sonntag, 1974; Porschen and Spaulding, 1974). Heptose and KDO are present in the lipopolysaccharide (Hofstad, 1974). *Fusobacterium mortiferum* is the only member of the genus *Fusobacterium* to have a mixture of mesolanthionine and diaminopimelic acids at an equimolar ration in its cell wall. Some strains are susceptible to cephalothin, cefazolin, cefoxitin, and lincomycin (Finegold, 1977). Resistant to 3 µg/ml erythromycin; susceptible to 12 µg/ml chloramphenicol, 2 U/ml penicillin G, and 6 µg/ml tetracycline (broth disk method).

Source: blood and various human clinical specimens, intestinal tract and feces; and one from irradiated mice.

DNA G+C content (*mol%*): 26–28 (T_m).

Type strain: ATCC 25557, CCUG 14475.

Sequence accession no. (*16S rRNA gene*): AJ867032.

Fusobacterium naviforme (Jungano 1909) Moore and Holdeman 1970, 45[AL] [*Bacillus naviformis* Jungano 1909, 123; *Ristella naviformis* (Jungano 1909) Prévot 1938, 291]

na.vi.for'me. L. n. *navis* ship; L. neut. suff. *forme* in shape of; N.L. neut. adj. *naviforme* in the shape of a ship.

Cells from glucose broth cultures are 4–10 µm that are often concave. Cells in old cultures have a beaded appearance. Surface colonies are punctiform to 2.0 mm in diameter, circular entire, low convex, gray-white, translucent with mottled appearance when viewed by obliquely transmitted light. Glucose broth cultures are lightly turbid with smooth to clumpy sediment and final pH of 5.5–6.4. Produces DNase (Porschen and Sonntag, 1974). No phosphatase is detected (Porschen and Spaulding, 1974). Heptose and KDO are present in the lipopolysaccharide (Hofstad, 1974). All strains produce indole from glucose-peptone medium (Gharbia and H.N. Shah, 1989) The cell walls contain *meso*-lanthionine as the major diaminopimelic acid. Some strains are resistant to 3 µg/ml erythromycin. Susceptible to 12 µg/ml chloramphenicol, 1.6 µg/ml clindamycin, 2 U/ml penicillin G, and 6 µg/ml tetracycline.

Source: the large intestine of a laboratory rat. Other strains have been isolated from the human gingival sulcus, from various human clinical specimens, and the bovine rumen.

DNA G+C content (*mol%*): 32–33 (T_m).

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Type strain: ATCC 25832, CCUG 50052, NCTC 13121.

Sequence accession no. (*16S rRNA gene*): not available.

Fusobacterium necrogenes (Weinberg, Nativelle and Prévot 1937) Moore and Holdeman 1970, 45[AL] [*Bacillus necrogenes* Weinberg, Nativelle and Prévot 1937, 681; *Spherophorus necrogenes* (*sic*) (Weinberg, Nativelle and Prévot 1937) Prévot 1938, 298]

ne.cro'ge.nes. Gr. n. *necros* the dead; N.L. suff. *-genes* (from Gr. v. *gennaô* to produce) producing; N.L. adj. *necrogenes* dead-producing, here necrosis-producing.

This description is based on our study of the type and three phenotypically similar strains. Cells from 1-d-old glucose broth cultures are extremely pleomorphic with coccoid cells about 0.3–0.8 μm and thin filamentous forms 0.2–0.8 μm in diameter and up to 20 μm in length. Cells in older cultures are somewhat more uniform, irregularly staining rods 0.7–0.8 × 1.5–4.0 μm. Surface colonies on horse blood agar are minute to 0.5 mm in diameter, circular, flat to low convex, entire, translucent, white, smooth, and shiny, and colonies are surrounded by zones of β-hemolysis. Glucose broth cultures are moderately turbid with pH of 5.7–6.0 Glutamine decarboxylase negative (Terada et al., 1976). Heptose and KDO present in the lipopolysaccharide (Hofstad, 1974).

Source: isolated by Kawamura (1926) from necrotic abscess of a chicken. Barnes strain EB/D/1/4a was isolated from cecal contents of a duck (Barnes and Impey, 1968). Other strains have been isolated from human feces.

DNA G+C content (*mol%*): 27–28 (T_m).

Type strain: ATCC 25556, CCUG 4949, NCTC 10723.

Sequence accession no. (*16S rRNA gene*): AJ867034.

Fusobacterium necrophorum (Flügge 1886) Moore and Holdeman 1969, 12[AL] [*Bacillus necrophorus* Flügge 1886, 273; *Fusiformis necrophorus* (Flügge) Topley and Wilson 1929, 299; *Sphaerophorus necrophorus* (Flügge) Prévot 1938, 298]

ne.cro'pho.rum. Gr. n. *necros* the dead; Gr. v. *phoreô* to bear; N.L. neut. adj. *necrophorum* dead producing, here necrosis producing.

Cells in glucose broth cultures are 0.5–0.7 μm in diameter with swellings up to 1.8 μm. The ends of the cells may be round or tapered. Cell length ranges from coccoid bodies to filaments over 1.00 μm. Filamentous forms with granular inclusions are more common in broth, while bacilli are more common in older cultures and growth on agar. Surface colonies on blood agar are 1–2 mm in diameter; circular with scalloped to erose edges, convex to umbonate, often with bumpy, ridged, or uneven surface; translucent to opaque, often with mosaic internal structure when viewed by transmitted light. Most strains produce either α- or β-hemolysis on rabbit blood agar. In general, the β-hemolytic strains are lipase positive (on egg yolk agar), and the α-hemolytic or nonhemolytic strains are lipase negative. No lecithinase is produced. Glucose broth cultures have a smooth, flocculent, granular, or stringy sediment and usually are turbid. The final pH of fructose and glucose cultures is 5.6–6.3. A few strains produce a pH of 5.8–5.9 in maltose medium.

Human rabbit and guinea pig red blood cells are agglutinated; bovine and ovine red blood cells are not (Simon, 1975). Dextran is not hydrolyzed (Holbrook and McMillan., 1977). No superoxide dismutase (Gregory et al., 1978) or lysine decarboxylase (Werner, 1974) is detected. Produce DNase (Porschen and Sonntag, 1974). No phosphatase is detected (Porschen and Spaulding, 1974). Heptose and KDO are present in the lipopolysaccharide (Hofstad, 1974).

Source: the natural cavities of humans and other animals and from clinical specimens (necrotic lesions, abscesses, and blood) of humans and other animals particularly liver abscesses and foot rot of cattle. For a review of natural and experimental pathogenicity, see Prévot et al. (1967) and Langworth (1977).

Source: bovine liver abscess; Fievez strain 2 3 58.

DNA G+C content (*mol%*): 31–34 (T_m) [chromatographic separation (Sebald, 1962)].

Type strain: ATCC 25286, CCUG 9994, CIP 104559, JCM 3718, VPI 2891.

Sequence accession no. (*16S rRNA gene*): AJ867039.

Further comments: two subspecies have been described among isolates of *Fusobacterium necrophorum*. These correspond to Biovar A and B.

Fusobacterium necrophorum subsp. necrophorum (Flügge 1886) Shinjo, Fujisawa and Mitsuoka 1991, 396[VP] (*Fusobacterium necrophorum* Moore and Holdeman 1969, 12)

ne.cro'pho.rum. Gr. n. *necros* the dead; Gr. v. *phoreô* to bear; N.L. neut.adj. *necrophorum* dead producing, here necrosis producing.

Previously known as *Fusobacterium necrophorum* biovar A. The subspecies name *Fusobacterium necrophorum* subsp. *necrophorum* (Flügge, 1886) Moore and Holdeman (1969) is automatically created by the valid publication of *Fusobacterium necrophorum* subsp. *funduliforme* (*ex* Hallé (1898) Shinjo et al.

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(1991) [Rule 40d (formerly Rule 46)] and was previously known as biovar B.

Type strain: ATCC 25286, CCUG 9994, CIP 104559, JCM 3718, VPI 2891.

Sequence accession no. (*16S rRNA gene*): AJ867039.

Fusobacterium necrophorum subsp. funduliforme (*ex* Hallé 1898) Shinjo, Fujisawa and Mitsuoka 1991, 396[VP] ("*Sphaerophorus funduliformis*" Hallé 1898)

fun.du.li.for'me. L. n. *fundulus* sausage; L. neut. suff. *forme* in shape of; N.L. neut. adj. *funduliforme* sausage-shaped.

Type strain: Fn524, ATCC 51357, CCUG 42162, CIP 104859, DSM 19678, JCM 3724.

Sequence accession no. (*16S rRNA gene*): AM905356.

Fusobacterium perfoetens (Tissier 1905) Moore and Holdeman 1973, 72[AL] [*Coccobacillus perfoetens* Tissier 1905, 110; *Ristella perfoetens* (Tissier 1905) Prévot 1938, 291; *Sphaerophorus perfoetens* (Tissier, 1905) Sebald 1962, 149]

per.foe'tens. L. prep. *per* very; L. part. adj. *foetens* stinking; N.L. part. adj. *perfoetens* very stinking.

Description is from Prévot et al. (1967), Weinberg et al. (1937), van Assche and Wilssens (1977). Cells from glucose cultures are 0.6–0.8 × 0.8–1.0 μm, oval, never elongated, occurring singly, in pairs, in chains of no more than three cells, or in masses. No flagella or capsule. Colonies in deep agar (2 d) are 1 mm in diameter and lenticular. Surface colonies on blood agar are 1–2 mm in diameter, circular with an entire edge, convex to raised, grayish white, translucent, smooth, and nonhemolytic on horse blood. Colonies of some strains are slightly umbonate with diffuse edges and a slightly mottled or granular appearance.

Growth in glucose broth is rapid. Cultures are turbid with a fine to ropy sediment and a pH of 5.6. Gas and a fetid odor are produced. Galactose is weakly fermented (Van Assche and Wilssens, 1977). Growth is enhanced in media containing fructose, glucose, mannose, sucrose, and trehalose. The pH of cultures in these media ranges from 5.6 to 5.95. Produces CO_2 and NH_3. Major amounts of lactic acid produced from PYG cultures (Van Assche and Wilssens, 1977) no lactic acid detected in PYG cultures of the type strain.

Inhibited by 0.001% polymyxin. Resistant to 0.001% brilliant green. Optimum temperature is 37°C; good growth occurs at 45°C, poor growth at 25–30°C. Survives up to 24 h of exposure to air.

Source: isolated by Tissier in 1900 from an infant with diarrhea and in 1905 from nursing infants. Strain CC1 isolated in 1947 by Prévot from the cecum of a horse and studied by Sebald (1962) has been lost. Van Assche and Wilssens (1977) studied 6 isolates from the feces of a 2-week-old pig, one of which has been designated the neotype strain.

DNA G+C content (*mol%*): 28–30 (T_m) (Sebald, 1962; Van Assche and Wilssens, 1977).

Type strain: ATCC 29250.

Sequence accession no. (*16S rRNA gene*): M58684.

Fusobacterium peridonticum Slots, Potts and Mashimo 1984, 270[VP] (Effective publication: Slots, Potts and Mashimo 1983, 963.)

pe.ri.o'don.ti.cum. Gr. prep. *peri* around; Gr. n. *odous*, *ontos* tooth; L. neut. suff. *-icum* suffix used with the sense of pertaining to; N.L. neutr. adj. *periodonticum* pertaining to periodonte.

Description is from Slots et al. (1982). Obligately anaerobic, nonmotile, nonsporeforming, Gram-stain-negative rod. Mean cell size on blood agar is 0.5–1.0 × 4.0–7.0 μm, but filaments longer than 100 μm are often present. Cells had pointed to slightly rounded ends and generally occurred singly or in pairs lying end to end. After anerobic incubation for 2–3 d on blood agar, the colonies measured from 2.0 to 3.0 mm in diameter, were circular, convex, entire or slightly scalloped at the edge, slightly rough, and granular and opaque. Glucose broths were turbid, with flocculent or stringy sediment. Indole is produced, hydrogen sulfide formed from cysteine, litmus milk reduced, hippurate hydrolyzed, and fructose, galactose, and glucose fermented. Growth is inhibited by 1 μg/ml chloramphenicol, penicillin G, and tetracycline and was unaffected by 5 μg/ml erthryomycin and vancomycin. Addition of 2% oxgall inhibited the growth.

DNA G+C content (*mol%*): 28 (T_m).

Type strain: EK1-15, ATCC 33693, CCUG 14345, JCM 12991.

Sequence accession no. (*16S rRNA gene*): X55405.

Fusobacterium russii (Hauduroy, Ehringer, Urbain, Guillot and Magrou 1937) Moore and Holdeman 1970, 45[AL] (*Bacteroides russii* Hauduroy, Ehringer, Urbain, Guillot and Magrou 1937, 73)

rus'si.i. N.L. masc. gen. n. *russii* of Russ, named after V. Russ, the bacteriologist who first cultured this organism.

Cells in glucose broth are 1.5–4.0 μm in length and may form thin filaments 10–15 μm in length. Surface colonies on horse blood agar are 0.5–1 mm in diameter, circular, smooth, shiny, entire, convex, and translucent. The type strain is β-hemolytic on horse blood agar. Glucose-peptone broth cultures are turbid, often with stringy sediment and

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have a final pH of 5.9–6.1. Produces DNase and phosphatase (Porschen and Sonntag, 1974; Porschen and Spaulding, 1974). Heptose and KDO are present in the lipopolysaccharide (Hofstad, 1974). The major dibasic amino acid is *meso*-lanthionine. Strains do not ferment any carbohydrates nor produce lactate threonine, but phosphatase activities were reported (Table 1). Susceptible to 12 μg/ml chloramphenicol, 1.6 μg/ml clindamycin, and 2 U/ml penicillin G. Some strains are resistant to 6 μg/ml tetracycline.

Source: isolated by Russ (1905) from perianal abscess. Also isolated from infections of cats, including actinomycosis of cats and from human and animal feces.

DNA G+C content (*mol%*): 31 (T_m).

Type strain: ATCC 25533.

Sequence accession no. (*16S rRNA gene*): X55409.

Fusobacterium simiae Slots and Potts 1982, 193[VP]

sim'i.ae. L. fem. n. *simia* monkey; L. gen. n. *simiae* of the monkey.

Cellular morphology resembles that of *Fusobacterium nucleatum* (*q.v.*, species 1). Fructose and glucose are fermented (pH of 5.5–5.6). Twenty-seven other substrates tested are not fermented. Indole and lipase are produced; hippurate is hydrolyzed. Grows in media containing 2% oxgall. From glucose, butyrate is the major fermentation product; major amounts of acetate and small amounts or propionate, lactate, and succinate also are produced. Neither hydrogen nor gas is detected. Lactate and threonine are converted to propionate. The type strain has 48% DNA homology with the type and one other strain of *Fusobacterium nucleatum* and 9% homology with the type strain of *Fusobacterium necrophorum*.

Source: the mouth of the stump-tailed macaque (*Macaca arctoides*).

DNA G+C content (*mol%*): 27–28 (T_m).

Type strain: 7511 R2-13, ATCC 33568, CCUG 16798 (Slots and Potts 7511 R2–13).

Sequence accession no. (*16S rRNA gene*): X55407.

Fusobacterium ulcerans Adriaans and Shah 1988, 447[VP]

ul'ce. rans. L. part. adj. *ulcerans* making sore, causing to ulcerate, referring to the source of isolation.

Gram-stain-negative, nonsporeforming, obligately anaerobic, rod-shaped 0.5–4.5 μm long. Most strains consist of long cells with pointed ends. Surface colonies on anaerobic blood agar plates are 2–3 mm in diameter, circular, entire, domed to low convex, cream colored, and nonhemolytic. A second morphological type consists of rod-shaped organisms that are 0.8–4.5 × 0.2 μm and have a large round swelling (diameter 0.5 μm) in the center of the cell. Surface colonies on blood agar plates are 1–2 mm in diameter, circular, entire, and low convex. They are translucent or white in color, butyrous, and nonhemolytic. The optimal temperature for growth is 37°C with colonies appearing in 36–48 h. Culture in brain heart infusion broth produce generalized turbidity after incubation for 3–4 d.

Fermentation of peptone-yeast-glucose broth produces large amounts of butyric acid and small to moderate amounts of acetic, propionic, lactic, and succinic acids. All strains convert threonine to propionic acid. Both colonial types produce gas when grown in deep culture. No H_2S is detected. Acid is produced from glucose, and some strains ferment glucose and mannose. No acid is produced from fructose, lactose, sucrose, maltose, arabinose, raffinose, or rhamnose. Indole, catalase, lecithinase, urease, lipase, and oxidase are not produced. Esculin and starch are not hydrolyzed. Nitrate is reduced by all strains.

All strains are susceptible to penicillin and phosphamycin and are resistant to rifampin.

DNA G+C content (*mol%*): 29.2–29.5 (T_m).

Type strain: ATCC 49185, CCUG 50053, NCTC 12111.

Sequence accession no. (*16S rRNA gene*): AJ867037.

Fusobacterium varium (Eggerth and Gagnon 1933) Moore and Holdeman 1969, 12[AL] [*Bacteroides varius* Eggerth and Gagnon 1933, 409; *Sphaerophorus varius* (Eggerth and Gagnon 1933) Prévot 1938, 299]

va'ri.um. L. neut. adj. *varium* diverse, different, various.

Cells from glucose broth cultures are pleomorphic, coccoid and rod-shaped, and stain unevenly. Cells are 0.3–0.7 × 0.7–2.0 μm and occur singly and in pairs. Surface colonies on blood agar are punctiform to 1 mm in diameter, circular with entire edges, flat to low convex, translucent, usually with gray-white centers, and colorless edges. Glucose broth cultures are turbid with smooth sediment and a final pH of 5.3–5.7. Dextran is not hydrolyzed (Holbrook and McMillan., 1977). Produces lysine decarboxylase (Werner, 1974). Produces DNase (Porschen and Sonntag, 1974). No phosphate detected (Porschen and Spaulding, 1974). Heptose and KDO are present in the lipopolysaccharide (Hofstad, 1974).

A temperate lysogenic bacteriophage is harbored by the reference ATCC 27725 which causes a lytic response in other strains of *Fusobacterium varium* including the reference strain

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ATCC 8501. The phage is activated by incubation at 45°C and UV exposure (Gharbia, personal observation)

Source: human feces, purulent infections of humans (upper respiratory tract, surgical wounds, peritonitis), cecal contents of mice, intestinal contents of *Blatta orientalis* (roach), posterior intestinal tract of *Recticulitermes lucifugus* (termite) and vaginal swab of chinchilla.

DNA G+C content (*mol%*): 29 (T_m) (Sebald, 1962).

Type strain: ATCC 8501, CCUG 4858, NCTC 10560.

Sequence accession no. (*16S rRNA gene*): AJ867036.

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